EXP-8A

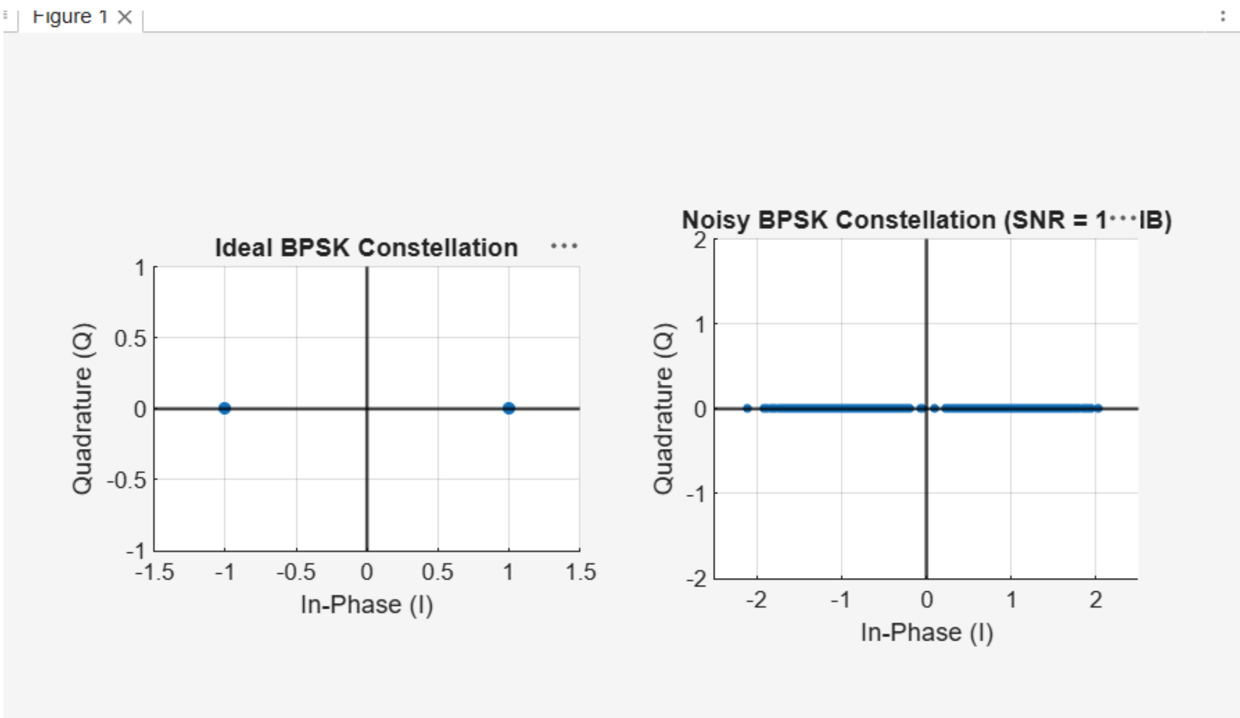
CODE:8A

```
clc;
clear;
close all;
%% Objective 1: Generate BPSK Symbols
N = 2000;                    % Number of bits
data = randi([0 1], N, 1);   % Random binary data
% BPSK Mapping: 0 -> -1, 1 -> +1
bpsk_symbols = 2*data - 1;
%% Objective 2: Plot BPSK Signal Constellation
figure;
subplot(1,2,1);
scatter(real(bpsk_symbols), imag(bpsk_symbols), 30, 'filled');
hold on;
xline(0, 'k', 'LineWidth', 1.5);
yline(0, 'k', 'LineWidth', 1.5);
grid on;
axis equal;
xlim([-1.5 1.5]);
ylim([-1 1]);

xlabel('In-Phase (I)');
ylabel('Quadrature (Q)');
title('Ideal BPSK Constellation');
```

```
%% Objective 3: Add AWGN Noise
SNR = 10;                    % SNR in dB
rx_signal = awgn(bpsk_symbols, SNR, 'measured');
subplot(1,2,2);
scatter(real(rx_signal), imag(rx_signal), 15, 'filled');
hold on;
xline(0, 'k', 'LineWidth', 1.5);
yline(0, 'k', 'LineWidth', 1.5);
grid on;
axis equal;
xlim([-2.5 2.5]);
ylim([-2 2]);
xlabel('In-Phase (I)');
ylabel('Quadrature (Q)');
title(['Noisy BPSK Constellation (SNR = ', num2str(SNR), ' dB)']);
%% Objective 4: Symbol Detection
detected_bits = real(rx_signal) > 0;
%% Objective 5: Calculate Bit Errors and BER
num_errors = sum(data ~= detected_bits);
BER = num_errors / N;
fprintf('Number of Transmitted Bits : %d\n', N);
fprintf('Number of Bit Errors       : %d\n', num_errors);
fprintf('Bit Error Rate (BER)       : %.5f\n', BER);
```

OUTPUT:



CODE:8B

```
clc;
clear;
close all;
%% Objective 1: Generate QPSK Symbols
N = 2000;                     % Number of bits
data = randi([0 1], N, 1);    % Random binary sequence

% Ensure even number of bits
if mod(N,2) ~= 0
```

```
    data = [data; 0];
    N = N + 1;
end
% Group bits into pairs
bit_pairs = reshape(data, 2, [])';
%% QPSK Mapping
qpsk_symbols = zeros(size(bit_pairs,1),1);
for k = 1:size(bit_pairs,1)
    if isequal(bit_pairs(k,:), [0 0])
        qpsk_symbols(k) = (1 + 1j)/sqrt(2);
    elseif isequal(bit_pairs(k,:), [0 1])
        qpsk_symbols(k) = (-1 + 1j)/sqrt(2);
    elseif isequal(bit_pairs(k,:), [1 1])
        qpsk_symbols(k) = (-1 - 1j)/sqrt(2);
    else
        qpsk_symbols(k) = (1 - 1j)/sqrt(2);
    end
end
%% Objective 2: Plot Ideal QPSK Constellation
figure;
scatter(real(qpsk_symbols), imag(qpsk_symbols), 30, 'filled');
hold on;
xline(0, 'k', 'LineWidth', 1.5);
yline(0, 'k', 'LineWidth', 1.5);
grid on;
axis equal;
xlim([-1.5 1.5]);
```

```
ylim([-1.5 1.5]);
xlabel('In-Phase (I)');
ylabel('Quadrature (Q)');
title('Ideal QPSK Constellation');
%% Objective 3: Add AWGN Noise
SNR = 10;                    % SNR in dB
rx_signal = awgn(qpsk_symbols, SNR, 'measured');
figure;
scatter(real(rx_signal), imag(rx_signal), 15, 'filled');
hold on;
xline(0, 'k', 'LineWidth', 1.5);
yline(0, 'k', 'LineWidth', 1.5);
grid on;
axis equal;
xlim([-2 2]);
ylim([-2 2]);
xlabel('In-Phase (I)');
ylabel('Quadrature (Q)');
title(['Noisy QPSK Constellation (SNR = ', num2str(SNR), ' dB)']);

%% Objective 4: Symbol Detection
detected_bits = zeros(N,1);

for k = 1:length(rx_signal)

  I = real(rx_signal(k));
  Q = imag(rx_signal(k))
```

```
    if I > 0 && Q > 0
        bits = [0; 0];
    elseif I < 0 && Q > 0
        bits = [0; 1];
    elseif I < 0 && Q < 0
        bits = [1; 1];
    else
        bits = [1; 0];
    end
    detected_bits(2*k-1:2*k) = bits;
end
%% Objective 5: BER Calculation
num_errors = sum(data ~= detected_bits);
BER = num_errors / N;

fprintf('\n');
fprintf('Number of Transmitted Bits : %d\n', N);
fprintf('Number of QPSK Symbols     : %d\n', length(qpsk_symbols));
fprintf('SNR                        : %d dB\n', SNR);
fprintf('Number of Bit Errors       : %d\n', num_errors);
fprintf('Bit Error Rate (BER)       : %.5f\n', BER);
```

OUTPUT:

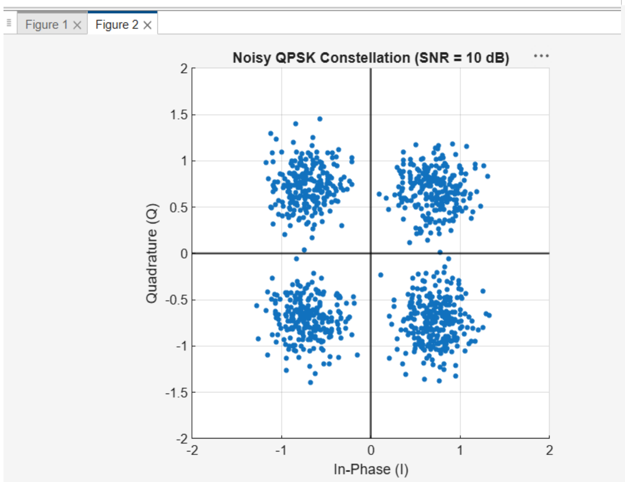


Figure 1
Figure 2
Noisy QPSK Constellation (SNR = 10 dB)
Quadrature (Q)
In-Phase (I)

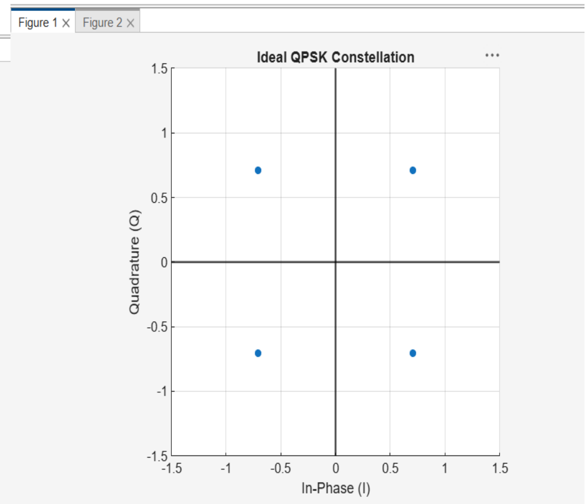


Figure 1
Figure 2
Ideal QPSK Constellation
Quadrature (Q)
In-Phase (I)